

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++ , SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.

Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.

Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

Accelerated provider onboarding and integration delivery by building RESTful Python backend services and TypeScript APIs that connected third-party clinical systems to live telehealth workflows, reducing manual configuration overhead across multiple customer deployments.

Improved diagnostic accuracy and reduced manual clinical review by integrating LLM-based inference pipelines (PyTorch, Llama) directly into provider-facing workflows via WebRTC, owning the full lifecycle from data ingestion to structured output on AWS SageMaker and Fargate.

Enabled faster, more reliable releases across production environments by owning YAML-based CI/CD configuration and cloud infrastructure automation, cutting environment drift and deployment cycle time.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).

Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

University Course Planning/Review Platform (Class Best): Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.